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Palaeomagnetic study of Oligocene (24–30 Ma) lava flows from the Kerguelen Archipelago (southern Indian Ocean): directional analysis and magnetostratigraphy

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Abstract

We report palaeomagnetic determinations in order to estimate the palaeosecular variation of the geomagnetic field for the upper Oligocene period from a southern site. We measured the anisotropy of magnetic susceptibility (AMS) of basalts from five sections to check the reliability of applying tectonic correction and concluded that no stratigraphic corrections were needed. A positive reversal test implies that secondary components were sufficiently removed from the characteristic remanent magnetization. Also, the sampling was sufficiently random to accurately reflect upper Oligocene palaeosecular variation. Calculations of the between-flow angular standard deviation (S_B) from the geographic pole show values between 21.2 and 21.8°. This palaeosecular variation estimate agrees with Model G for palaeosecular variation [J. Geophys. Res. 96 (B3) (1991) 3923], which is axisymmetric and assumes an equatorial symmetry of the geomagnetic field. The virtual geomagnetic poles (VGPs) from the Mont des Ruches section seem to exhibit two polarity changes: reverse–normal (R–N) first, and in more detail a second normal–reverse (N–R) one. Finally, assuming that no polarity chrons are missed by any of the sections, we compare the magnetostratigraphy we defined to the global geomagnetic polarity scale of [Geophys. J. Int. 129 (1) (1997) 176] using previously obtained radiometric dates to correlate the sections [J. Petrol. 39 (4) (1998) 711; Earth Planet. Sci. Lett. 174 (2000) 313; J. Petrol. 43 (7) (2002) 1341]. The precise age constraints furnished by these correlations allow an estimation of eruption rates from 0.30 to at least 1.31 km/my, with wide variations from section to section and an increase of eruption rate for younger sections. However, due to the non-uniqueness of the correlations, the age limits of the sections and consequently the extrusion rate estimations can vary. Moreover, assuming the Mont des Ruches section that has probably recorded all the magnetic polarity intervals, the Mont de la Rabouillère and Mont de la Tourmente intermediate VGP directions may indicate unrecorded polarity chrons. The eruption rates calculated can thus be overestimated.
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1. Introduction

Because there is no way to observe the composition and dynamics of the Earth's core directly, one indirect method is to study the Earth's magnetic field

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fluctuations recorded or observed at the Earth's surface. These fluctuations, which include secular variation, reversals, and excursions, are principally related to the fluid dynamics of the outer core. An accurate understanding of core dynamics can be obtained by the use of geomagnetic field modeling (see Dormy et al. (2000) for a review). However, these simulations require more experimental constraints to define the robust characteristics of the Earth's magnetic field against which they can be tested. Thus, numerous records more or less uniformally distributed at the Earth's surface for consecutive time windows on the geological time scale are needed (Merrill and McFadden, 1999). Unfortunately, owing to the non-uniform distribution of the continents, a sampling bias between the northern and southern hemisphere, particularly at high latitudes, affects the palaeomagnetic data currently available. Only few studies present volcanic sequences located in the southern hemisphere. A question that arises is whether the main palaeomagnetic field is equatorially symmetric, as used for some models of palaeosecular variation (McFadden et al., 1991), or whether it is more complicated as currently observed.

This report presents a palaeomagnetic study carried out on samples from five stratigraphic sections of Oligocene lava flows from the Kerguelen Archipelago (49.9°S, 70°E). Our objectives were to add new magnetic data in a poorly defined area, to prospect for flows recording transitions, and to check the symmetry of the geomagnetic field between the northern and southern hemisphere. Finally, by comparing the magnetostratigraphy of the five studied sections with previous geochemical studies and geochronological dating (Gautier et al., 1990; Nicolaysen et al., 2000; Frey et al., 2000; Doucet et al., 2002), we discuss the correlation between disparate stratigraphic sections and more precisely constrain calculated eruption rates.

2. Geological setting

The Kerguelen Archipelago is located in the southern Indian Ocean (49.9°S, 70°E) on the northern

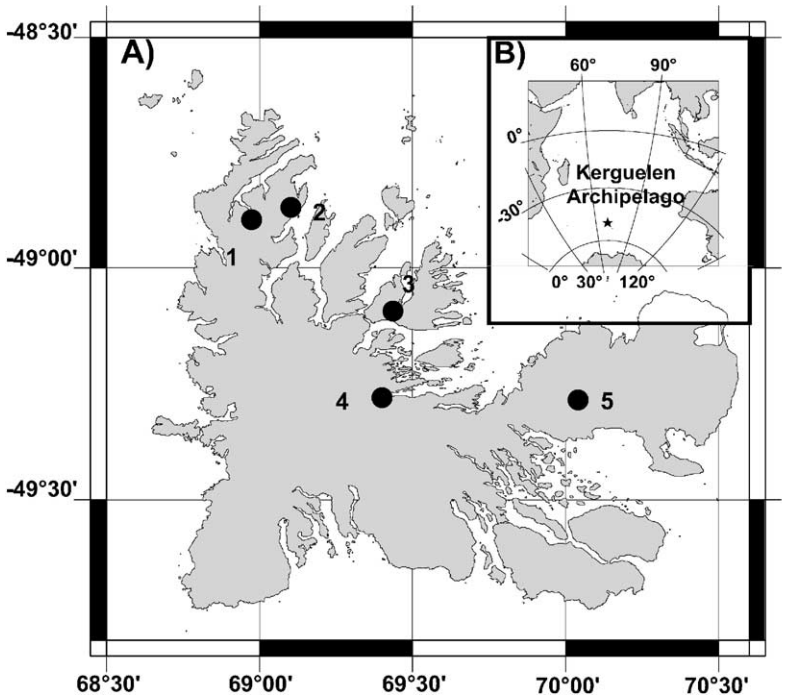


Fig. 1. (A) Locations of sampled sections: (1) Mont des Ruches: 18 flows, (2) Mont des Tempêtes: 20 flows, (3) Mont de la Rabouillère: 19 flows, (4) Mont de la Tourmente: 33 flows and (5) Mont Amery: 8 flows. (B) Location of the Kerguelen Archipelago in the Indian Ocean.

part of the Kerguelen–Gaussberg Plateau (Fig. 1). This large oceanic plateau, with its conjugate Broken Ridge, is a manifestation of intensive Cretaceous (119–95 Ma) volcanism due to the Kerguelen hotspot (Duncan, 2002 and references therein). The Kerguelen Archipelago represents the continuation of this hotspot volcanism for the last 30 Ma (Yang et al., 1998; Weis et al., 1998; Nicolaysen et al., 2000), but some magmatic contribution from the Southeast Indian Ridge (SEIR) during the early Cenozoic is not excluded (Doucet et al., 2002). More than 85% of the Archipelago is composed of basalt flows which form valley walls (400–900 m high) exposed by glacial erosion. Intrusions (gabbro, granite, and syenite) from 1 to 15 km diameter also occur. Isotopic chemistry links the origin of the plutons to Kerguelen plume melts (Giret, 1990; Weis and Giret, 1994).

3. Palaeomagnetic sampling

In recent geochemical study of seven volcanic sections located principally in the northern part of the Kerguelen Archipelago, Nicolaysen et al. (2000) and Doucet et al. (2002) determined $^{40}Ar/^{39}Ar$ isochron ages on acid leached groundmass. To benefit from these available dates, the strategy was to sample the same sections. Then, we looked for the thickest studied sections, recovering the maximum number of successive and minimally-altered lava flows. We also searched for sampling areas not affected by significant tectonism or by subsequent intrusions. Based on geologic, geochemical, and logistical criteria, we decided to target the five sections presented in Fig. 1. In addition to a previous palaeomagnetic study (Camps et al., in preparation), this allowed us to completely sample the Oligocene basalts exposed on the Archipelago. Except for the Mont de la Tourmente section, we were unable to sample the previously-dated sections in exactly the same locations. For the Mont des Ruches section, syenite and dyke intrusions close to the original geochimical section led us to sample the opposite valley wall (800 m westward), assuming the two sections are correlated. For the Mont des Tempêtes section, we were not able to cross a major river to reach the 3 km distant Mont Fontaine section. Here again, we sampled the opposite valley wall assuming the lavas correlate between the two sections. For the Mont Rabouillère section, we preferred to take a nearby section (400 m away) which was less altered and undisturbed by dykes. In this case, there is no doubt of the lateral continuity between the two sections. Finally, for the Mont Amery section, we chose to continue the previous magnetic sampling of Henry and Plessard (1997), starting at the next flow upward, rather than to resample the dated Mont Crozier section (about 2 km away) which is more difficult to reach.

In all, almost a hundred lava flows with seven samples per flow on average were drilled using a gas-powered drill, and the samples were oriented with both solar sightings and magnetic compass plus a clinometer. Because of the particular field trip conditions encountered on Kerguelen, the possibility to return to the same place is always uncertain. For this reason, we decided to sample each successive lava flow, in order to recover any existing magnetic transitional directions.

4. Rock magnetic properties

4.1. Low-field susceptibility versus temperature

4.1.1. Experimental procedure

We performed continuous measurements of the low-field (100 μT) magnetic susceptibility versus temperature (*KT* curves) during heating and cooling cycles to characterize the magnetic mineralogy of samples and to estimate their thermal stability. We used the method of Prévot et al. (1983) to estimate the mean Curie temperature of magnetic minerals revealed by a reversible fall in the *KT* curves. The measurements were performed in vacuum ($<10^{-2}$ mbar) using a modified Bartington susceptibility meter MS2 associated to a home made furnace. The rock sample is a cylindrical core of $\approx 1\,cm^3$. One sample per flow for the Mont des Ruches, Mont des Tempêtes and Mont Rabouillère sections was heated to its Curie temperature with a heating and cooling rate of 7°/min. Unfortunately, some technical problems did not permit analysis of all the flows, but the shape of the curves we obtained are independent of the corresponding section. The two treated flows from the Mont de la Tourmente section make us confident to expect that the two remaining sections will qualitatively provide similar results.

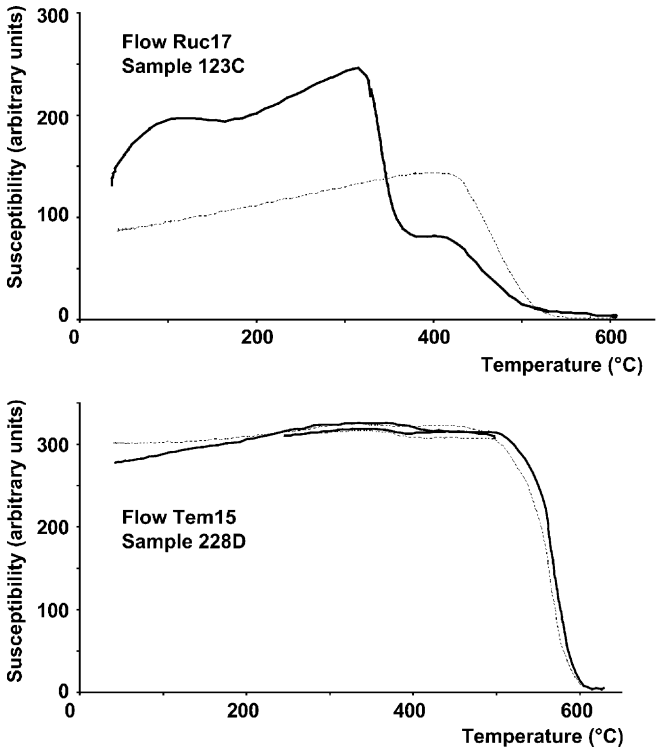


Fig. 2. Low-field susceptibility versus temperature curves displaying the two typical behaviours observed for the Kerguelen basalts. Bold (dashed) lines correspond to heating (cooling) phases.

4.1.2. Results and interpretation

The primary magnetic carrier is presumed to be a titanomagnetite ($Fe_{3-x}Ti_xO_4$) with an x value of 0.6–0.7 as generally expected for basaltic lava flows. Two distinct types of *KT* behaviour were observed. The first case, sample 123C in Fig. 2 illustrates the irreversible and complex thermomagnetic behaviour observed for almost 60% of the samples. This sample presents three susceptibility falls at ≈100–150°C, near 350°C and finally around 400°C. We suggest that the first susceptibility decrease corresponds to the original titanomagnetites. However, titanomagnetites are supposed to stay stable during heating under vacuum and should be observed in the cooling curve. Therefore, it is not excluded that the first fall indicate slightly oxidized titanomaghemites, resulting from low temperature oxydation of the original titanomagnetites. Heating–cooling cycles at various intermediate temperatures performed on some other specimens lead us to interpret the second fall as due to the presence of titanomaghemites with higher degree of oxidation or different mineral size but also at the same time to dissociation effects of the metastable titanomaghemites present in the sample. Because inversion of titanomaghemites is an irreversible phenomenon, the third susceptibility fall might not exist before heating in the laboratory. Titanomaghemites are known to invert in two stages during heating–cooling cycles, with a first unstable inversion product which produces the third susceptibility decrease near 400°C, and a final stable inversion product which depends on the degree of oxidation of the starting titanomaghemite (Özdemir, 1987). These final products usually consist of a titanomagnetite–ilmenite assemblage, or haematite and pseudobrookite for the most highly oxidized titanomaghemite (Özdemir, 1987). No matter which is the degree of oxidation because as it is often observed, haematite is reduced under vacuum to yield a Ti-poor titanomagnetite (∼400°C). In case of total oxidation, the susceptibility measured during cooling

might only reflect this titanomagnetite. Thus, whatever the degree of oxidation considered, the shoulder in the susceptibility curve observed during cooling might relate the presence of a stable titanomagnetite with a $460 \pm 25^{\circ}$C Curie temperature. All samples showing this complex behaviour yield qualitatively the same *KT*-curve with some differences due probably to varying degrees of oxidation of the original titanomaghemite.

For the second case, sample 228D in Fig. 2, the reversibility of the curve indicates that the primary mineral phase is a non-maghemitized titanomagnetite with a mean Curie temperature of $560 \pm 15^{\circ}$C. This low-Ti titanomagnetite quite probably results from high temperature oxyexsolution during the flow emplacement of the original titanomagnetite.

Unfortunately, the two behaviours describe above cannot be related to macroscopic flow characteristics as for example, the flow thickness, the color of the samples or the presence of phenocrysts. Thus, a complete study of each flow is needed to detect the thermally stable second case which could correspond to samples well suited for palaeointensity determinations (Plenier et al., in preparation).

4.2. Viscosity index

4.2.1. Experimental procedure

Before magnetic cleaning, the viscosity index ν (Thellier and Thellier, 1944; Prévot, 1981) was first determined on the entire collection in order to estimate the viscous remanent magnetization (VRM) possibly acquired by the samples since the beginning of the Brunhes polarity epoch. We measured the specimens after 2 weeks in the ambient magnetic field oriented along the z core axis and again, after 2 weeks storage in a zero field. The difference between these two measurements yield the VRM acquired during 15 days in the present geomagnetic field (Thellier and Thellier, 1944). We calculate the viscosity index ν using

$$\nu = \frac{\Delta_z}{\sqrt{x_0^2 + y_0^2 + z_0^2}} \times 100$$

where Δ_z represents the difference in the magnetization intensity parallel to the magnetizing field between the two measurements, and x_0, y_0 and z_0 represent the cartesian components in core coordinate of the magnetization after 2 weeks storage in a field-free space. The viscosity index, expressed as a percentage, corresponds approximately to 25% of the VRM acquired in situ since the last reversal of polarity (780 kyr) (Prévot, 1981).

4.2.2. Results and interpretations

The mean geometric viscosity indices ($\bar{\nu}$) for each flow are reported in Table 1. Calculated for the whole collection, $\bar{\nu}_{all}$ is relatively high (10.3%) compared to the mean value of 6.1% for subaerial volcanic rocks found by Prévot (1981). Only 33% of the specimens have a viscosity index <5 or $<10\%$ for intermediate polarity. Surprisingly, the mean geometric viscosity index calculated for the reverse polarity units alone ($\bar{\nu}_R$) is lower than for normal ones ($\bar{\nu}_N$), 5.8 and 16.4% respectively, which contradicts normal expectations (Prévot, 1981).

However, the high normal polarity $\bar{\nu}_N$ indices and the difference in $\bar{\nu}$ values for normal and reverse polarity flows are due to the Mont de la Tourmente section only. As observed in Fig. 3A, the four other localities present more expected values of ν, with more than 50% of the samples displaying a viscosity index lower than 10%. For the Mont de la Tourmente section, only 20% of the samples display a similar ν value. Moreover, we can observe that almost 15% of the samples from this section present viscosity indices greater than 100%. Fig. 3B and C illustrate that these huge values are associated to samples providing low magnetization intensity, after two weeks storage in a shielded room, combined with larger viscous component, after 15 days under the present geomagnetic field.

The particular behaviour of the Mont de la Tourmente samples is quite difficult to explain. According to Prévot (1981), VRMs in basalts can arise in two ways. VRMs hard with respect to alternating fields (AFs) demagnetization are carried by single domain magnetite grains with sizes near the superparamagnetism threshold, resulting from high-temperature oxidation of originally homogeneous titanomagnetite. AF-soft VRMs are carried by multidomain grains of low Curie point homogeneous titanomagnetite. Because the ratio of VRM on the total remanent magnetization is supposed to increase with the grain size of magnetic mineral, our results may indicate a larger magnetic grain size for the Mont de la Tourmente lava flows than for classical subaerial lavas. As already

Table 1
Cleaned average directions of magnetization of the lava flows

Flow	n/N	Inc	Dec	α_{95}	κ	P_{lat}	P_{long}	Polar	ν%	J_{20}	Alt
Mont des Ruches section (48.87°S, 68.91°E)											
Ruc18	7/7	72.7	175.7	4.3	197.7	−80.5	82.8	R	9.13	0.37	230
Ruc17	7/7	65.6	192.6	5.6	116.4	−81.6	336.2	R	9.85	0.53	220
Ruc16	4/4	79.9	183.3	6.5	201.6	−68.4	65.9	R	2.48	0.50	180
Ruc15(1)	7/7	74.5	159.2	3.9	244.8	−73.1	105.3	R	4.6	1.17	160
Ruc14(1)	7/7	76.4	149.5	4.5	183.8	−67.7	104.5	R	5.8	1.15	150
Ruc13	7/7	64.0	237.6	11.1	30.5	−51.8	356.7	R	32.45	0.33	140
Ruc12	7/7	64.0	69.4	6.0	102.3	−22.2	113.8	T	65.81	0.45	120
Ruc11	7/7	81.3	144.1	8.0	58.2	−61.2	89.8	R	0.23	1.73	100
Ruc10	7/7	78.3	92.7	5.3	130.5	−45.1	101.7	R	1.25	1.08	90
Ruc9	11/11	−14.4	15.0	8.1	32.4	46.6	90.8	N	35.88	1.10	85
Ruc8	8/8	−60.8	15.9	2.5	474.4	76.8	132.5	N	17.11	0.59	80
Ruc7	6/7	−62.0	345.1	4.9	188.2	78.3	1.9	N	21.2	0.85	70
Ruc6(2)	8/8	−68.7	358.2	4.1	180.1	86.6	268.0	N	2.57	3.69	60
Ruc5(2)	7/7	−59.1	0.6	13.1	22.2	81.0	71.9	N	9.36	1.80	55
Ruc4(2)	7/7	−66.2	10.1	5.5	120.6	83.3	160.3	N	4.45	2.92	50
Ruc3(2)	8/8	−70.6	356.6	3.9	203.2	83.7	267.0	N	4.28	5.37	40
Ruc2	7/7	−60.9	58.2	3.3	329.2	49.6	171.8	N	5.21	1.08	25
Ruc1	8/8	52.5	195.1	5.3	108.6	−70.6	290.0	R	4.11	1.28	10
Mont des Tempêtes section (48.88°S, 69.11°E)											
Tem20	7/8	67.4	165.3	5.1	139.9	−80.4	145.5	R	25.52	0.16	246
Tem19	6/7	73.4	142.8	7.2	88.1	−66.2	119.3	R	72.64.	0.20	224
Tem18	8/8	60.7	177.0	3.5	248.7	−82.5	231.6	R	2.73	9.47	172
Tem17	7/7	62.0	186.0	4.0	224.8	−83.0	287.7	R	3.36	13.40	160
Tem16	5/7	64.2	178.1	4.2	328.6	−86.8	224.6	R	5.57	3.62	152
Tem15(3)	4/4	64.5	188.1	2.8	1109.8	−84.0	317.3	R	4.04	5.80	145
Tem14(3)	7/7	57.1	190.3	7.9	59.3	−76.6	286.6	R	2.06	15.10	140
Tem13	7/7	55.5	180.5	5.0	146.0	−77.2	250.9	R	2.99	1.76	130
Tem12(4)	7/7	59.8	175.0	5.2	136.0	−81.1	223.9	R	10.48	1.23	120
Tem11(4)	7/7	58.5	185.1	4.3	199.5	−79.7	271.7	R	3.91	0.45	110
Tem10(4)	7/7	60.6	188.0	7.0	75.6	−80.8	289.7	R	2.56	0.75	105
Tem9	7/7	58.7	186.7	7.2	70.5	−79.4	278.5	R	11.64	0.47	100
Tem8	6/7	88.4	106.2	4.1	269.4	−49.7	73.9	R	47.41	0.17	90
Tem7(5)	7/7	77.3	219.9	4.3	193.6	−63.4	33.1	R	5.28	0.22	80
Tem6(5)	5/5	77.1	221.2	2.4	979.6	−63.0	31.9	R	4.94	0.53	75
Tem5	5/5	80.9	251.2	4.4	308.3	−51.5	41.5	R	5.93	0.39	68
Tem4	7/7	78.3	232.9	3.2	360.7	−58.0	34.0	R	9.00	0.33	60
Tem3(6)	7/7	74.4	203.1	14.4	18.5	−69.2	42.4	R	21.44	1.38	50
Tem2(6)	7/7	64.6	215.7	4.4	193.1	−66.1	347.0	R	41.99	1.10	30
Tem1	7/7	−67.3	356.6	2.8	464.7	87.5	309.2	N	5.87	3.20	5
Mont de la Tourmente section (49.30°S, 69.39°E)											
Tou1	7/7	−69.9	358.0	4.9	150.0	85.3	264.1	N	13.21	1.15	580
Tou2	7/7	−47.7	3.8	4.1	222.0	69.3	78.9	N	2.97	0.57	525
Tou3	7/7	−49.6	6.2	4.9	152.6	70.6	85.6	N	4.63	8.33	520
Tou4	7/7	−69.6	358.3	5.3	131.4	85.8	263.4	N	23.08	0.72	510
Tou5	9/9	−56.1	332.0	3.9	174.9	66.1	0.8	N	53.48	1.10	505
Tou6(7)	7/7	−62.6	356.4	5.5	123.6	84.1	43.2	N	8.82	1.15	495
Tou7(7)	4/4	−63.9	346.2	11.1	69.0	80.0	355.9	N	37.93	0.86	490
Tou8(7)	7/7	−64.3	359.6	6.8	80.5	86.8	64.5	N	110.17	0.29	480
Tou9	7/7	−76.6	131.0	5.8	108.3	30.0	227.4	T	54.67	1.24	465
Tou10(8)	7/7	−65.0	11.9	3.5	306.5	81.7	147.6	N	9.47	1.20	430
Tou11(8)	6/6	−72.1	6.5	4.3	247.0	81.3	225.5	N	20.26	2.12	420
Tou12	7/7	−52.6	22.4	7.6	63.9	71.2	120.8	N	52.44	0.80	410
Tou13	7/7	−55.6	34.4	6.1	98.4	61.8	144.6	N	6.35	1.78	390

Table 1 (*Continued*)

Flow	n/N	Inc	Dec	α_{95}	κ	P_{lat}	P_{long}	Polar	ν%	J_{20}	Alt
Tou14	3/4	−77.8	304.0	16.2	58.8	57.2	286.8	N	65.6	0.41	380
Tou15[9]	7/7	−66.6	23.9	5.7	112.1	74.5	167.8	N	37.13	0.95	370
Tou16[9]	7/7	−73.7	33.3	2.8	462.8	68.4	200.6	N	26.48	0.39	360
Tou17	6/7	−39.8	285.7	42.5	3.4	27.0	335.5	T	172.63	5.97	340
Tou18	7/7	−59.6	323.8	2.0	883.5	63.2	344.8	N	26.72	0.29	330
Tou19	7/7	−75.5	14.2	5.5	121.7	74.6	224.4	N	4.74	2.51	320
Tou20	7/7	−72.0	6.6	3.7	267.6	79.1	225.5	N	18.80	1.33	305
Tou21	7/7	−56.7	2.6	7.0	75.8	77.8	79.3	N	148.60	0.18	290
Tou22	6/7	−84.4	8.7	10.7	39.9	52.6	256.5	N	96.79	0.30	280
Tou23	7/7	−67.8	24.6	5.0	144.3	74.2	174.1	N	11.33	3.05	250
Tou24	7/7	−69.5	10.8	5.5	121.6	82.2	193.5	N	10.99	0.43	240
Tou25	7/7	−79.3	16.4	7.0	76.0	68.5	233.6	N	18.11	1.60	220
Tou26	8/8	−64.6	18.2	6.1	83.0	77.5	153.2	N	59.41	0.42	210
Tou27	7/7	−65.4	26.7	2.6	551.0	72.3	163.8	N	24.75	0.55	200
Tou28	7/7	−67.2	35.4	7.2	71.0	67.3	174.6	N	26.07	1.68	190
Tou29	5/5	−52.0	100.7	8.7	78.1	17.9	189.0	T	22.44	2.12	180
Tou30	7/7	−77.6	41.2	2.6	551.1	63.1	213.6	N	18.62	2.13	160
Tou31	7/7	−60.0	6.2	5.7	112.8	80.5	99.1	N	30.38	0.24	135
Tou32	7/7	−66.1	38.5	4.3	199.1	64.9	172.3	N	12.51	0.34	120
Tou33	7/7	−59.6	34.4	3.9	243.9	64.3	152.4	N	54.88	0.94	110
Mont Amery section (49.29°S, 70.06°E)											
Ame8	7/7	−77.7	350.8	3.2	347.2	72.2	262.1	N	2.33	2.56	345
Ame7	7/7	−74.9	326.0	6.6	83.4	67.5	294.0	N	32.09	0.14	340
Ame6	7/7	−57.2	37.1	4.1	220.5	61.1	150.8	N	14.51	2.48	330
Ame5	7/7	−60.8	29.5	5.4	125.0	68.2	151.2	N	13.28	0.69	280
Ame4	7/7	−52.9	9.6	5.3	129.3	72.6	97.9	N	18.73	0.87	270
Ame3	7/7	−75.2	9.4	3.7	265.3	76.1	231.5	N	19.14	0.56	250
Ame2	7/7	−60.4	22.3	6.4	89.7	72.5	141.5	N	7.05	0.12	225
Ame1	7/7	−60.5	1.7	4.3	195.5	82.1	79.4	N	17.45	0.63	200
Mont Rabouillère section (49.09°S, 69.44°E)											
Rab19	7/7	−72.54	2.63	2.8	471.0	81.1	240.6	N	8.8	3.72	280
Rab18	7/7	−78.14	2.63	4.3	196.0	71.6	246.3	N	2.4	1.49	270
Rab17	7/7	−63.17	109.68	4.4	187.0	22.1	203.4	T	9.0	0.30	250
Rab16	8/8	−73.07	354.00	2.0	922.0	79.8	267.4	N	9.3	1.99	230
Rab15	7/7	75.27	173.30	4.1	220.0	−76.1	82.8	R	13.8	0.33	190
Rab14	7/7	69.52	173.53	3.7	269.0	−84.1	111.1	R	10.7	0.20	170
Rab13	7/7	48.40	134.43	2.3	714.0	−50.4	172.0	R	4.8	1.06	160
Rab12	7/7	68.19	140.23	4.2	207.0	−64.8	138.7	R	1.8	0.84	150
Rab11	7/7	56.93	193.60	3.7	266.0	−74.9	295.0	R	2.6	1.85	130
Rab10	6/7	48.70	178.36	1.5	2014.0	−70.5	245.2	R	1.5	1.67	120
Rab9	7/7	71.72	177.11	2.7	494.0	−82.3	81.8	R	23.4	0.22	100
Rab8	7/7	68.22	176.29	2.8	476.0	−86.6	112.7	R	18.3	0.18	80
Rab7	7/7	69.85	198.78	1.9	1017.0	−77.4	8.3	R	11.2	0.59	60
Rab6	6/7	59.00	185.50	4.1	273.0	−80.0	274.2	R	4.9	3.41	45
Rab5	7/7	67.96	180.88	3.1	375.0	−87.9	55.7	R	9.9	1.06	30
Rab4	8/8	54.53	188.27	1.7	1073.0	−74.7	276.0	R	5.7	3.63	20
Rab3[10]	8/8	81.02	160.99	3.1	319.0	−65.1	82.7	R	0.9	2.67	14
Rab2[10]	7/7	81.71	152.27	4.0	234.0	−62.5	85.7	R	2.5	1.34	8
Rab1[10]	7/7	78.38	145.84	2.5	600.0	−64.7	99.4	R	7.8	0.94	2

The superscript ($i = 1$–10) indicates flows which can be grouped together (see discussion); n/N the number of samples used in the analysis/total number of samples collected; Inc and Dec the mean inclination, positive downward, and the declination east of north, respectively; α_{95} the 95% confidence envelope for the average direction; κ the precision parameter of Fisher distribution; P_{lat}/P_{long} the latitude/longitude of VGP position, respectively. The polarity of the flow (Polar) is normal (N) if $P_{lat} > 45°$, reverse (R) if $P_{lat} < -45°$, and intermediate (T) for the other cases. ν% is the geometric mean viscosity index (Thellier and Thellier, 1944); J_{20} the geometric mean remanence intensity after cleaning at 20 mT (for AF demagnetized samples only); Alt is the altitude of the flow in meter.

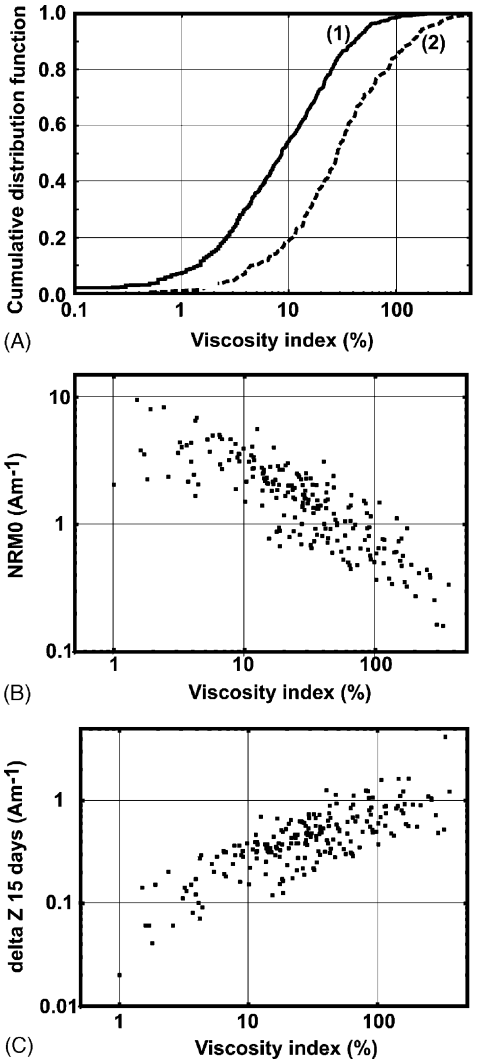


Fig. 3. (A) Cumulative distribution function of the viscosity indices (ν) from (1) Mont des Ruches, Mont des Tempêtes, Mont de la Rabouillère, and Mont Amery sections combined, (2) the Mont de la Tourmente section. (B) Magnetization intensity after 2 weeks storage in a shielded room (NRM0) as a function of ν. (C) Magnetization intensity of the viscous component acquired during 15 days under the geomagnetic field (Δ_z 15 days) as a function of ν.

deduced from the *KT* analysis, both low Curie temperature (first case) and high temperature oxidation (second case) are observed in Kerguelen samples. We will demonstrate that the VRM carried by the samples is soft (removed at 20 mT), favoring the hypothesis of larger multidomain grains of low Curie point titanomagnetites to explain the high normal polarity values.

5. Palaeodirection determinations

5.1. Tectonic corrections

No significant dip has been observed for four of our sections. On the contrary, the whole Mont de la Tourmente section presents an eastward dip of 15°. According to the geological map (Nougier, 1970), only east–west trending fractures are present in the neighborhood of this section, and there is no direct evidence for a tilting of the entire section after flow emplacement.

5.1.1. Anisotropy of magnetic susceptibility (AMS)

It is possible to discriminate paleotopography and associated flow direction from post-emplacement tectonic tilting by comparing the flowing direction, determined using AMS and the present dip direction. We measured the AMS of the samples from the Mont de la Tourmente section to know if the tectonic corrections were needed. In order to recover the initial flow direction (Cañón Tapia et al., 1995), all the samples, with regular standard shape ($n = 221$ of 244), were analyzed using the KLY2 kappabridge in the IPG Paris Palaeomagnetic Laboratory at Saint Maur. In order to eliminate statistically indistinct values, we filtered the individual data using both F statistics and the three sigma limit on the A and B factors as described by Cañón Tapia et al. (1995). Thus, 33 samples were eliminated. The confidence regions for the mean direction of each axis were calculated using linear perturbation analysis (Hext, 1963; Jelinek, 1978; Lienert, 1991), bootstrap method (Constable and Tauxe, 1990) and the bivariate extension of Fisher's statistics (Henry and Le Goff, 1995). A difficulty in AMS study of volcanic flows is the possible occurrence of permutation of maximum-intermediate susceptibility axis. Such permutations of the principal susceptibility axis are sometimes observed in samples from a same flow, possibly giving wrong mean directions (Cañón Tapia et al., 1996). We therefore determined the density contour of the maximum (K_{max}), intermediate (K_{int}), and minimum (K_{min}) principal susceptibility axis to look for possible discrepancies

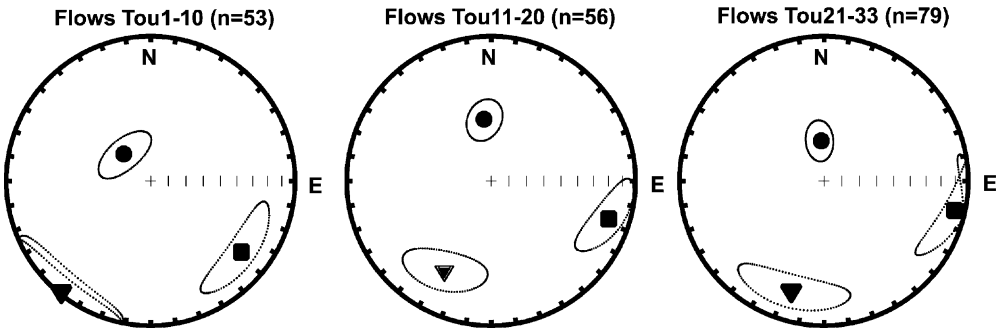


Fig. 4. Mean principal susceptibility axis of three consecutive parts form the Mont de la Tourmente section and their 95% confidence ellipses (Constable and Tauxe, 1990), square K_{max}, triangle K_{int} and circle K_{min}.

between orientation of the axis clusters and mean directions.

AMS measurement of fewer than 10 specimens per flow yields quite large statistical uncertainty of the susceptibility tensor (Cañón Tapia et al., 1997). Although all our samples were drilled from the lower part of each flow, the sampling strategy was not designed for AMS analysis. Because only seven samples were taken per flow, AMS mean directions per flow are of weak significance, showing large uncertainties. However, the aim of our study was not to constrain the flowing direction of each flow, because it can be related to the local flowing disturbances, but to determine the mean flowing direction. The low significance of the data per flow does not allow determination of this mean flowing direction from running direction in the different flows. The lack of observed discordances between the flows suggests that the flowing conditions were relatively constant for the whole section, and we therefore merged all the AMS data.

To verify that such a merging is justified, a comparison of the mean data was made dividing the section in three subsets. The coherent results obtained in these parts (Fig. 4 and Table 2) clearly show the constant flowing conditions during emplacement of the whole section. The three statistical methods used to define the mean directions and their associated confidence zones, provided approximately the same

Table 2
Directions of the mean principal susceptibilities and sizes of the confidence regions around them for the Mont de la Tourmente section

		$\bar{s}$	$\delta\bar{s}$	Dec	Inc	ζ	d_ζ	i_ζ	η	d_η	i_η
Flows 1–10 ($N = 53$)	Max	0.336	0.0005	128.2	21.0	9.6	295.8	68.5	26.0	36.5	4.2
	Int	0.334	0.0005	219.1	2.4	14.9	73.9	87.1	27.9	309.1	1.7
	Min	0.330	0.0006	315.3	68.8	8.7	140.7	21.1	17.7	50.0	1.8
Flows 11–20 ($N = 56$)	Max	0.336	0.0005	107.9	16.0	9.6	355.7	52.8	21.8	208.4	32.5
	Int	0.334	0.0003	207.4	30.2	12.4	346.4	52.3	22.0	105.0	20.3
	Min	0.330	0.0004	353.7	55.0	9.3	104.8	14.1	12.6	203.5	31.3
Flows 21–33 ($N = 79$)	Max	0.336	0.0003	103.0	7.9	7.6	359.4	59.4	27.4	197.5	29.3
	Int	0.334	0.0003	196.1	21.3	11.3	8.0	68.5	27.4	105.1	2.8
	Min	0.330	0.0004	353.8	67.2	7.6	85.0	0.5	11.8	175.2	22.8
Whole section ($N = 188$)	Max	0.336	0.0003	113.6	16.1	5.7	351.0	61.9	14.0	210.4	22.4
	Int	0.334	0.0002	209.3	18.9	7.5	348.5	65.7	14.0	114.1	14.7
	Min	0.330	0.0003	345.9	64.8	5.7	107.7	13.9	7.5	203.0	20.6

N is the number of sample per data-set; $\bar{s}$ the eigenvalue of normalized mean susceptibility tensor; $\delta\bar{s}$ corresponds to the 95% confidence region for $\bar{s}$; Dec and Inc the declination and the inclination of the mean susceptibility axis, respectively; ζ and η the semiangles of minor and major axis of the 95% confidence ellipse, respectively; $d_{\zeta,\eta}$ the declination of minor, major axis direction; $i_{\zeta,\eta}$ the inclination of minor, major axis direction.

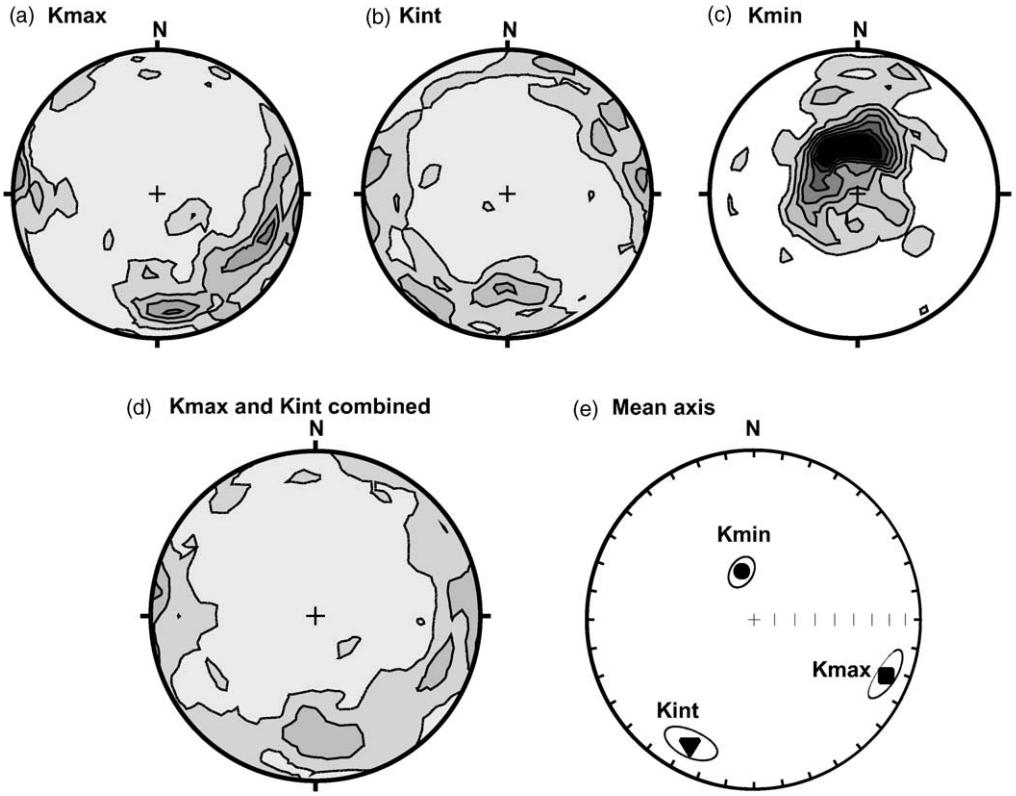


Fig. 5. (a–d) Density diagrams of the principal susceptibility axis K_{max} (a), K_{int} (b), K_{min} (c) and K_{max}–K_{int} (d) combined for the Mont de la Tourmente section ($n = 188$). Mean principal axis (e) and their 95% confidence ellipses (Constable and Tauxe, 1990), square K_{max}, triangle K_{int} and circle K_{min}.

results, for each part as for the whole data. Fig. 5a–d displays for all the samples the K_{max}, K_{int}, K_{min} and K_{max}–K_{int} density diagrams, respectively. K_{max}, K_{int} and K_{max}–K_{int} density contours present bipolar diagrams. For K_{max}, the two maxima of density are ESE and S oriented. K_{int} own a comparable S trend but the second maximum is E in this case. The combination of the K_{max} and K_{int} leads to decrease the density contrasts, with a maximum of W–E to WNW–ESE main direction and another one, weaker, of S azimuth. The K_{min} density diagram displays a well defined mean direction, with a maximum of density strongly dipping northwestward.

5.1.2. Discussion

The bipolarity of the K_{max} and K_{int} density diagrams clearly show that K_{max}–K_{int} axis permutation is important in this section. However, one of the maximum density of K_{max} and the maximum density of K_{min} perfectly agree with the mean K_{max} and K_{min}, determined using the other statistical approaches. This indicates that permutations do not significantly affects the mean tensor directions. The consistency of the mean directions for the different parts of the section and the relatively small 95% confidence areas for the whole section yield a strong and coherent signal among the section. The mean tensor directions can be therefore used for the determination of the flowing direction.The interpretation of these data is unfortunately questionable. In fact, because of axis permutation, K_{max} or K_{int} could represent the flowing direction. Moreover, both mean K_{max} and mean K_{int} present a significant inclination. K_{min} is not in the vertical plane including K_{max} or K_{int}. On the one hand, K_{max} corresponds to the strongest maximum density and

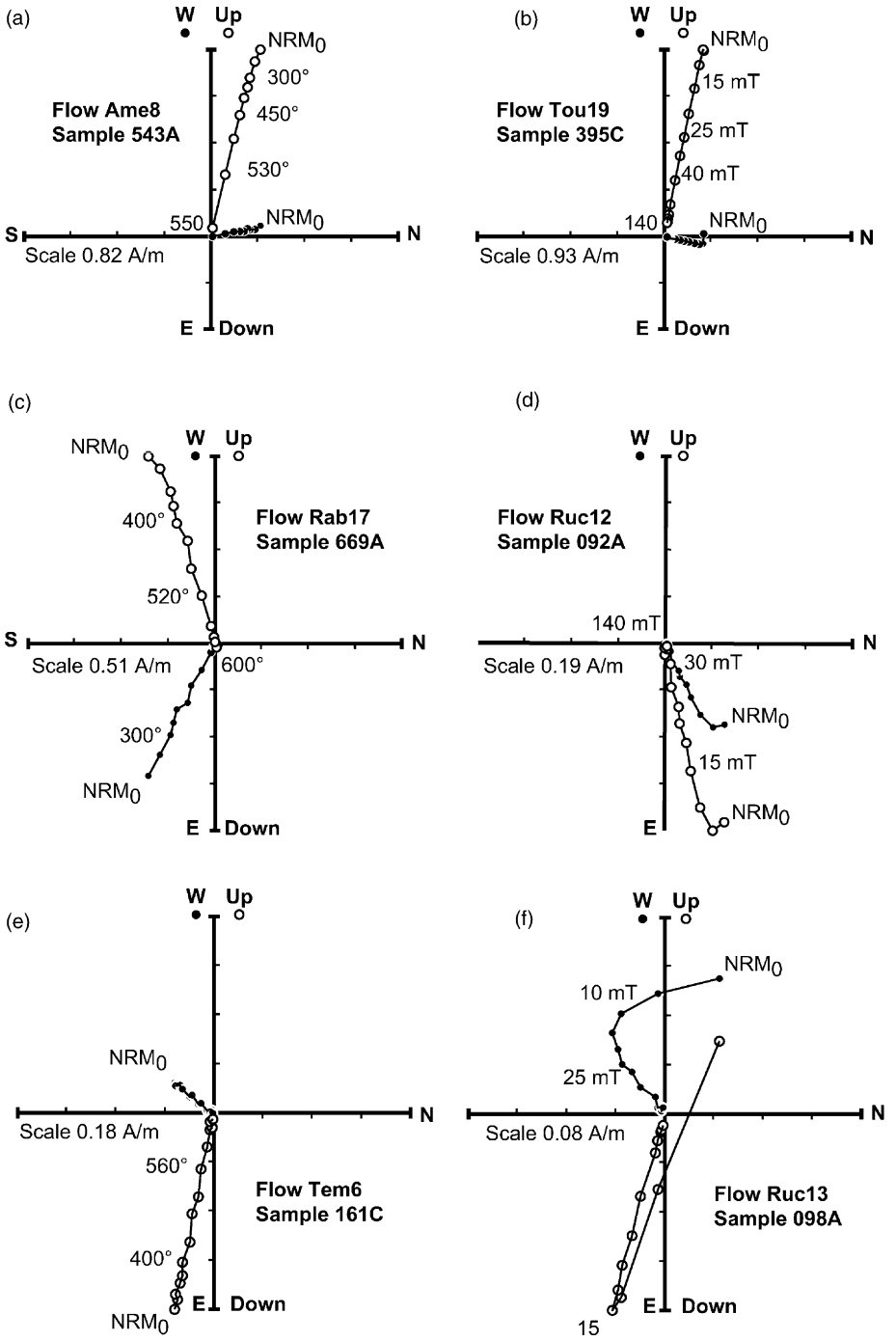


Fig. 6. Orthogonal plots for normal reversed and intermediate flows for differently demagnetized samples in geographic coordinates. The black (white) points correspond to the horizontal (vertical) plane.

should be a good candidate. In this case, the flowing direction agrees with the present dip direction and probably no tilting occurred after lava emplacement. On the other hand, if K_{int} represents the flowing direction, the flows were tilted after emplacement. Untilting leads to horizontal K_{max}, and to K_{min} and K_{int} in a same vertical plane. No decisive argument yields choice between these two possibilities. The VGPs from the five sections (see Table 1) are more axially distributed without tectonic corrections on the Mont de la Tourmente section. This argues for a eastward dip of topographic origin before the lava flowing. AMS data being compatible with a lack of tilting after lava emplacement, we chosen to not apply dip correction to the paleomagnetic data from the Mont de la Tourmente section.

5.2. Palaeomagnetic cleaning

We performed demagnetizations of the samples in two steps. First, two pilot samples per flow were stepwise demagnetized, one in AF using a laboratory-built AF demagnetizer (up to 140 mT) and the other thermally in a non-inductive PYROX furnace with a residual magnetic field of less than 20 nT. Remanent magnetizations were measured using a JR-5A spinner magnetometer after each demagnetization step (7–16). The behaviour of the pilot specimens guided our choice of the best demagnetization procedure for the remaining samples from the same flow. When the two demagnetization methods yield almost the same results, we preferred to proceed with the thermal treatment because the separation of components is usually more efficient for volcanic rocks. Moreover, the thermal demagnetizations reveal the spectrum of remanence unblocking temperatures, used later to define the demagnetization steps of a pilot palaeointensity experiment (Plenier et al., in preparation).

5.3. Results

For almost all of the samples, the individual characteristic remanent magnetization (ChRM) was clearly isolated. Nearly all samples are contaminated by a small secondary component, oriented along the present field, which is removed entirely below 20 mT or 350°C. The softness of this secondary component indicates a viscous origin. The relatively high viscosity indices we found, especially for the Mont de la Tourmente section, support this interpretation. Considering a limit of $\pm 45^\circ$ for the latitude of the virtual geomagnetic poles (VGPs) during stable polarity, examples of orthogonal vector plots for normal, reverse, and intermediate polarity are presented in Fig. 6. We determined the individual ChRM directions by means of the least-squares method (Kirschvink, 1980) using at least three consecutive demagnetization steps. If the individual ChRM could not be isolated or if the sample was possibly misoriented (as indicated by field notes and by a direction significantly different than the other samples from the same flow) the sample was excluded from further analysis. This is the case for 11 samples on the 684 analysed. The determinations are of sufficiently good quality since the maximum angular deviation (MAD) is lower than 2° for 80% of the samples. Moreover, only seven flows have an $\alpha_{95} > 10^\circ$, four of them being of intermediate polarity. The directional results are presented in Table 1. Of 98 lava flows, 51% are normal polarity, 44% are reversed polarity, and 5% display intermediate directions. Except for the mont des Ruches section (18 flows), where the normal and reverse polarity flows are equally represented, the percentage of reversed units varies a lot. For the Mont des Tempêtes and the Mont Rabouillère sections (20 and 19 flows, respectively), it is greater than 75%, whereas it is zero for the Mont de la Tourmente and the Mont Amery sections (33 and 8 flows, respectively).

6. Discussion

6.1. Magnetostratigraphy and extrusion rates

The polarity intervals measured in the five Kerguelen Archipelago sections are shown as a function of elevation in Fig. 7. We propose to discuss the correlations between these magnetostratigraphy informations and the reference polarity time scale (Huestis and Acton, 1997) using the absolute age determinations obtained on the corresponding previously dated sections (Yang et al., 1998; Nicolaysen et al., 2000; Doucet et al., 2002), and also the geochemical compositional variations of basalts. For example, the D group signature defined by Yang et al. (1998) in the upper part of the Mont de la Rabouillère section and observed by Doucet et al. (2002) for the lower part of

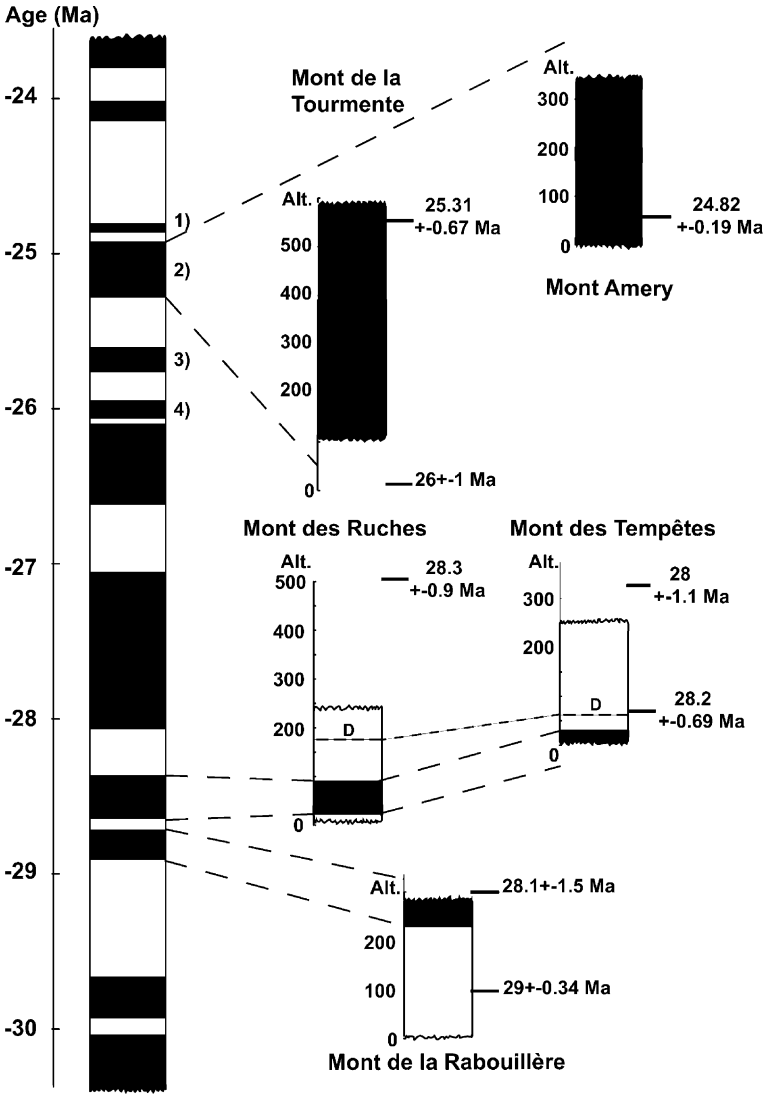


Fig. 7. Magnetostratigraphic correlations with the global geomagnetic polarity time scale (Huestis and Acton, 1997) interpreted using radiometric dates and geochemical observation (Yang et al., 1998; Nicolaysen et al., 2000; Doucet et al., 2002). We assume that all the polarity chrons are recorded at each section 1–4 are the possible correlations allowed for the Mont de la Tourmente section.

the Mont des Ruches and Mont des Tempêtes sections helps to place these sections stratigraphically with respect to each other. For the Mont Amery section, we sampled upward from the last units studied by Henry and Plessard (1997). We thus included their palaeomagnetic data, which describe the first 200 m of the section, to furnish a more complete magnetostratigraphy.

In order to check the relationship between the initiation of Kerguelen flood basalts and the conjunction of the Kerguelen Plume with the Southeast Indian Ridge, Nicolaysen et al. (2000) used lava accumulation rates to evaluate the location of the Kerguelen Archipelago when it began to form. The magnetostratigraphy correlated to the global geomagnetic polarity time scale allow to better constrain the age limits of each section

and then recalculate more accurate extrusion rates for each section. We calculated the extrusion rate (exr) as follows:

$$\mathrm{exr} = \frac{\Delta H}{\Delta t - \mathrm{ref}}$$

where ΔH is the analyzed stratigraphic thickness and $\Delta t - \mathrm{ref}$, the duration of the chrons the correlating lavas.

The correlations are made using the working hypothesis that all the polarity chrons spanning the emplacement of the volcanic sequence were recorded. This assumption seems reasonable because the volcanic activity of the Kerguelen hotspot is acknowledged to be vigourous. We did not see for example erosion surfaces between the lava flows, only thin sediment layers were sometimes observed. However, the volcanic activity of the Kerguelen plume may differ from place to place and the thickness of the sedimentary layers is not directly related to the time span between two successive flows, their absence does not necessary evidence a rapid succession. Without rigorous arguments, we tentatively considered that no hidden polarity chrons are missing from the volcanic sequences.

Because of the errors on age determinations, several correlations remain possible for each section. To overcome these uncertainties, we kept the nearest solution from the best radiometric dates available. For instance, the 26 ± 1 Ma date obtained at the base of the less constrained Mont de la Tourmente section is not directly related to a polarity boundary and the precision is not sufficiently accurate to allow a firm correlation. On the contrary, the 25.31 ± 0.67 Ma date at the top of the section is of quite good quality, but four correlations can be made considering the error bars (1–4 in Fig. 7). The correlation (2) corresponding to a normal polarity chron between 24.92 and 25.28 Ma is the nearest to 25.31 Ma and was kept as our preferred interpretation. One can argue that using dates determined on sections not sampled exactly at the same place can also introduce some bias, but as explained in the paleomagnetic sampling description Section 3, we are quite confident about the stratigraphic relation between the dated sections and the palaeomagnetic ones.

Concerning the extrusion rate calculations, depending on the interpretation made to correlate the sections to the global geomagnetic polarity scale, the results can vary significantly. For this reason, we will give both a lower and upper estimate. It is also important to notice that because the sections do not span the entire chrons, the time elapsed to produce each section is necesseraly shorter than the one used for the calculation. Therefore, both values of the extrusion rate are actually lower bounds of the true eruption rates. These limitations should be kept in mind to appreciate correctly the estimate proposed.

The interpretation, based on the nearest solution from the best radiometric dates, furnish the following constraints on the age limits. The Mont de la Rabouillère section accumulated between 29.7 and 28.7 Ma, the Mont des Ruches section between 28.7 and 28.1 Ma, the Mont des Tempêtes section between 28.6 and 28.1 Ma, the Mont Amery section between 24.9 and 24.8 Ma, and the Mont de la Tourmente section between 25.3 and 24.9 Ma. For the Mont de la Rabouillère, Mont des Ruches, and Mont des Tempêtes sections, we calculated an average extrusion rate of 0.33 km/my (0.70 km/my). But the well constrained normal polarity interval in between two reversed chrons observed on the Mont des Ruches section allows estimation of a more realistic eruption rate of 0.27 km/my (0.39 km/my) for this section. The upper limit of the D geochemical signature combined with the magnetic results indicate that the eruption rate for the Mont des Tempêtes section is almost two times lower than for the Mont des Ruches (between 0.16 and 0.23 km/my). For the youngest Mont de la Tourmente and Mont Amery sections, our first interpretation yields eruption rates of 1.31 km/my (8.55 km/my) and 6.27 km/my (0.96 km/my), respectively.

However, this last eruption rate seems unlikely because very high. Thus, we suggest that Mont de la Tourmente and Mont Amery sections erupted during the same normal polarity chron C7n.2n. This alternative interpretation still agrees with the radiometric dates within error. The age limits of the C7n.2n chron, between 25.3 and 24.9 Ma, yield a more acceptable eruption rate of 0.96 km/my for the Mont Amery section. This second interpretation is our preferred. Fig. 7 summarizes the retained correlations of each section with respect to the reference polarity timescale (Huestis and Acton, 1997).

Therefore, the Mont de la Tourmente and Mont Amery sections present smaller values, but within the incertitude interval, than the mean 1.6 ± 0.9 km/my

proposed by Nicolaysen et al. (2000) for the same sections. The eruption rates calculated by means of the magnetic remanence seem very different from section to section and confirm higher extrusion rates for youngest sections as supposed by Nicolaysen et al. (2000) from geochronological ages considerations. The major assumption used to obtain these conclusions is that all the polarity chrons are recorded at each section. If this is not the case, the extrusion rates are overestimated.

6.2. Intermediate directions

In the present study, we define arbitrarly a palaeomagnetic direction as intermediate if its corresponding VGP has a latitude between $\pm 45^{\circ}$. Intermediate directions can indicate polarity reversals as well as excursions of the geomagnetic field. Unfortunately, assuming that the magnetostratigraphy is complete at every sampling locality, the presented Kerguelen basalt sections did not record reversals because all the intermediate directions occurred within stable polarity intervals. Concerning the Mont des Ruches section, the preceding and following VGPs of the intermediate polarity flow possess a palaeopole latitude 30° or more away from the closest geographic pole. Thus it is possible to consider this intermediate direction as a rebound during a normal–reverse (N–R) polarity change. A similar remark can be made for the Ruc2 VGP location which might correspond to an intermediate position between the reverse–normal (R–N) polarity change. The successive VGPs from the Mont des Ruches section, connected together with great circles, are shown in Fig. 8. The fact that some VGPs can describe a reversal path lead us to consider, as a simplest interpretation, that all the reversals are recorded at least for the Mont des Ruches section. For the Mont de la Tourmente and Rabouillère sections, because any preceding and following VGPs are not deviated toward theses positions, the intermediate VGP seem isolated. Thus, they can either represent super palaeosecular variation, as well as excursions or true intermediate directions during a reversal not recorded. The assumption made to correlate these sections to a global geomagnetic polarity scale may be unsuitable.

6.3. Palaeosecular variation

We included in the present statistical analysis the palaeomagnetic results of Henry and Plessard (1997) for the Port Jeanne d'Arc, Port Couvreux and Mont Amery (Val Studer) sections in order to increase the number of upper Oligocene data. In a second step, we removed the transitional data presenting VGPs with latitude less than $\pm 45^{\circ}$ ($n = 9$). The palaeomagnetic directions of individual flow for this global dataset are displayed in Fig. 9. with their bootstrapped means and associated 95% confidence ellipses (Tauxe, 1998).

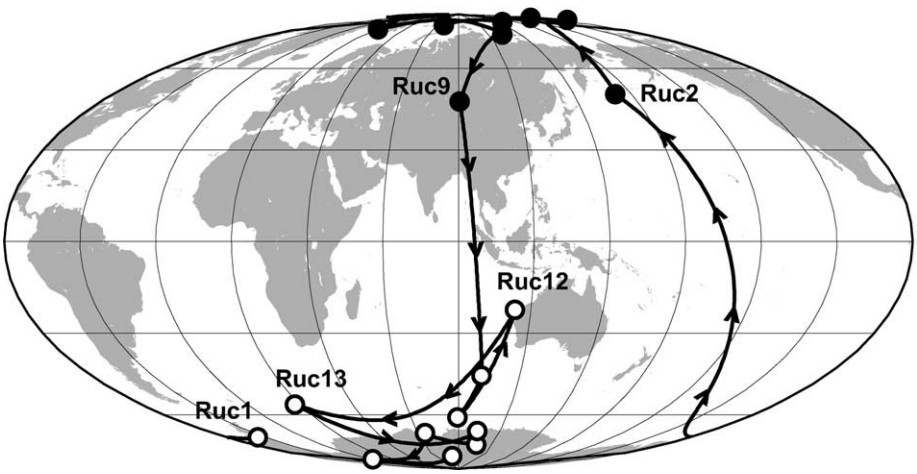


Fig. 8. Locations of VGPs from the Mont des Ruches section.

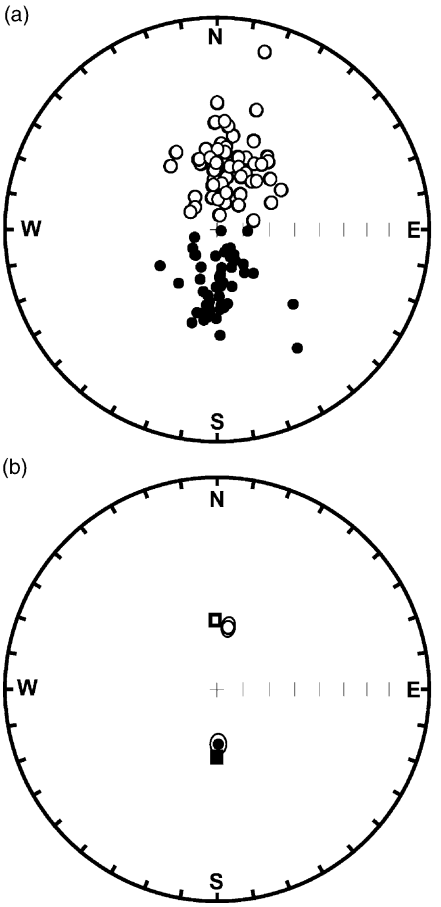


Fig. 9. (a) Palaeomagnetic directions of individual flow (Henry and Plessard (1997) and this study combined), and (b), their bootstrapped means with 95% confidence ellipses (Tauxe, 1998), after removing of the transitional data. The dipole field directions (squares) are given for comparison. In black (white) circles and squares are positive (negative) inclinations.

6.3.1. Bootstrap reversal test

In order to check the reliability of the data for palaeosecular variation study, we processed them by separating the mean directions into two polarity datasets and by inverting the normal polarity dataset to its antipode. Then we applied the bootstrap test for a common mean (Tauxe, 1998) for the two datasets in order to perform a reversal test. The statistical analysis of flow-average directions for these two dataset are presented in Table 3. Because the confidence intervals of the bootstrapped means for the two distributions overlap for each cartesian coordinate (Fig. 10), we

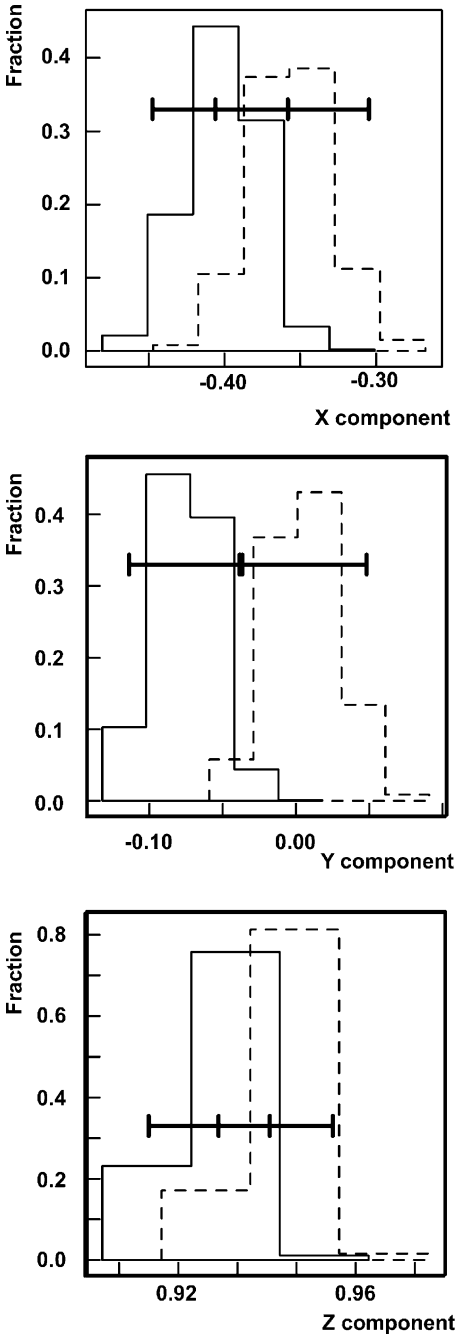


Fig. 10. Histograms of cartesian coordinates of the mean bootsrapped normal and reverse polarity directions and their 95% confidence intervals (Tauxe, 1998).

Table 3
Statistical analysis of flow-average direction

	Dec	Inc	N	R	κ	α_{95}
Ungrouped directions						
Normal	10.2	−65.6	60	58.0849	30.8	3.4
Reverse	178.7	68.9	49	47.5284	32.6	3.6
All combined	5.4	−67.2	109	105.4908	30.8	2.5
Grouped directions						
Normal	11.1	−65.3	53	51.1493	28.1	3.8
Reverse	177.2	68.5	41	39.7268	31.4	4.0
All combined	5.5	−66.9	94	90.7407	28.5	2.8

Dec, Inc are the mean declination and inclination, respectively; N the number of lava flow used in the analysis; R the vector sum of N unit vectors; κ the precision parameter of Fisher distribution; α_{95} the 95% confidence envelope for the average direction.

concluded that the two means are statistically identical at the 95% level of confidence. Such a positive reversal test implies further, first, that the individual ChRMs are not disturbed by secondary components, next that these ChRMs are primary TRMs, and finally that the palaeomagnetic directions reasonably average the secular variation during the upper Oligocene.

6.3.2. Flow grouping

A random sampling with sufficient time elapsed between successive field recordings is required to estimate the variability of the paleofield due to the secular variation (PSV). Because the present paleomagnetic sampling consists of successive lava flows in a vertical sequence, a risk to introduce a bias in the statistical analysis by giving a spurious weight to a single direction recorded by successive lava flows emplaced in a short time exists. To shed light on this problem, we use the quantitative bootstrap test for a common mean (Tauxe, 1998) to identify non-independent successive directions. When two or more succesive flows yield directions with a common mean, we used the field notes to check the geological evidences for elapsed time between the flows (conglomerates, red sediments, coal, etc.). The flows which cannot be distinguished using these two criteria are then grouped together as indicated in Table 1. Nonetheless, it is important to keep in mind that the palaeomagnetic field itself can stay stable during emplacement of successive lava flows with time elapsed between eruption long enough compared to the rate of secular variation (Love, 2000). In such a case, grouping the data will overestimate the variability of the paleofield. To quantify this potential problem, we estimated the palaeosecular variation using two extreme cases: with ($n = 94$) and without ($n = 109$) grouping of similar successive directions.

6.3.3. Estimation of the palaeosecular variation

We processed the combined data, by reversing the VGPs of reversed polarity. The total angular standard deviation (asd) is calculated using

$$S_T = (N-1)^{-1/2}\left(\sum_{i=1}^{N}\theta_i^2\right)^{1/2}$$

where N is the number of data and θ_i the angle between the ith VGP and the reference position. The total asd is then corrected for experimental error by removing the within-flow asd:

$$S_B^2 = S_T^2 - \frac{S_W^2}{\overline{n}}$$

where $\overline{n}$ is the mean number of samples per unit, $S_W \approx 81^\circ/\sqrt{\overline{k}}$ is the within-flow asd, and $\overline{k}$ the mean precision parameter. Results of these calculations are reported in Table 4. Because of the low number of

Table 4
Dispersion statistics of VGP

	VGPs (>45°)	Grouped VGPs (>45°)
N	109	94
Mean latitude	85.6	85.8
Mean longitude	202.0	197.7
Precision parameter K	15.5	14.6
α_{95}	3.6	4.0
Within flow asd (S_W)	5.00	4.98
Dispersion from mean VGP		
Total asd (S_T)	20.85	21.50
Between-flow asd (S_B)	19.35	19.95
Confidence limits (95%) on S_B	17.70–21.35	18.12–22.18
Dispersion from the Earth's rotation axis		
Total asd (S_T)	21.3	21.9
Between-flow asd (S_B)	21.2	21.8
Confidence limits (95%) on S_B	19.4–23.4	19.8–24.3

N is the number of flow for calculations. S_B , S_T, and S_W are the between, total, and within flow angular standard deviations, respectively. The 95% confidence limits on S_B are calculated using the table of 95% confidence limits for angular standard deviation (Cox, 1969).

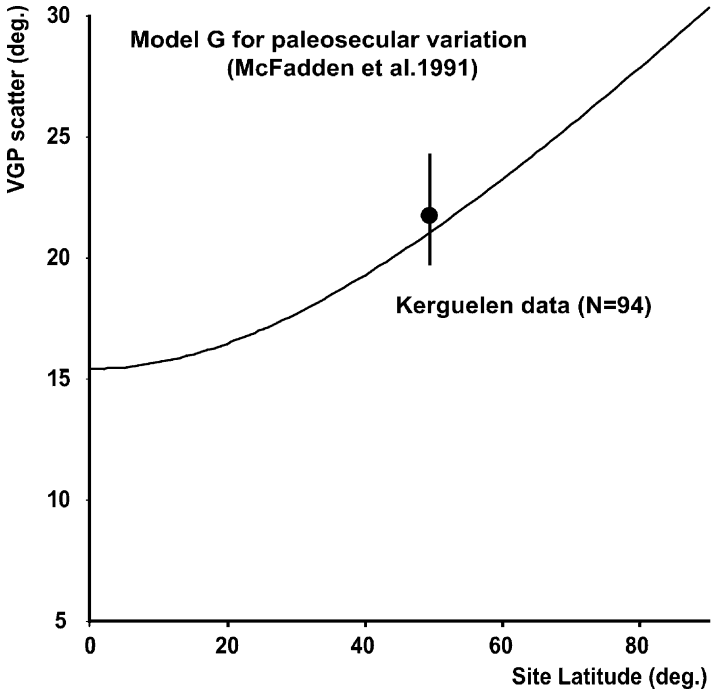


Fig. 11. VGP scatter and 95% confidence interval about spin axis of data ($n = 94$) from Henry and Plessard (1997) and the present study combined (with grouping of magnetically identical flows) compared to the Model G for palaeosecular variation (McFadden et al., 1991).

grouped flows, the two different data-sets considered do not change significantly the results. The real palaeosecular variation is estimated in the range of 19.4–24.3°, which corresponds to the lower and upper bounds of the two extreme cases.

6.3.4. Comparison with palaeosecular variation model

In Fig. 11, we report the VGP scatter and the associated 95% confidence interval about the spin axis ($n = 94$) to compare it with the Model G for palaeosecular variation (McFadden et al., 1991). We used this model because it is the only one which can be applied for the 24–30 Ma time window. Obviously, the data fit nicely the model. This substantiates the Model G of McFadden et al. (1991) for the approximation of the palaeosecular variation during the 22.5–45 Ma time interval. The equatorial symmetry of the main geomagnetic field assumed in PSV Model G is then coherent with our southern hemisphere dataset.

7. Conclusions

We present a paleomagnetic analysis of five lava sections from the Kerguelen Archipelago. We use the AMS technique to determine the initial flow direction and conclude that no tectonic corrections to sample orientations are needed for this study. The magnetic carriers have been identified using low-field susceptibility evolution curves and include: (1) original titanomagnetite or slightly oxidized titanomaghemite, (2) titanomaghemite with higher Curie temperature, more or less thermally stable due to low temperature oxidation at a different level, and (3) poor-Ti titanomagnetite produced by high temperature oxidation. Surprisingly, the calculated viscosity index of the Mont de la Tourmente section is larger than expected for subaerial lavas. After laboratory treatment, the primary remanent magnetization is well isolated, as shown by little within flow dispersion and a positive bootstrap reversal test. This test also assures that the Oligocene palaeosecular variation is well averaged

by the sampling. If we assume that no reversals are missed by the sections, and considering our interpretation of correlation to the Huestis and Acton (1997) global geomagnetic polarity scale, the magnetostratigraphy permits us to improve the dates of each section—between 29.7 and 28.7 Ma for The Mont de la Rabouillère section, 28.7 and 28.1 Ma for the Mont des Ruches section, 28.6 and 28.1 Ma for the Mont des Tempêtes section, and between 25.3 and 24.9 Ma for the Mont Amery and Mont de la Tourmente sections, and calculate more precisely the eruption rates. These extrusion rates –0.35 km/my for the Mont des Ruches section, 0.43 km/my for the Mont des Tempêtes, and lower bound values of 0.3, 0.96 and 1.31 km/my for the Mont de la Rabouillère, Mont Amery and the Mont de la Tourmente, respectively—present important variations from section to section with an increase for the younger sections. Thus, the assumption by Nicolaysen et al. (2000) of a constant eruption rate for the Eocene–Oligocene activity of the Kerguelen plume is an oversimplification. The intermediate data from the Mont des Ruches section is considered as a rebound during a normal to reverse polarity change. The VGP direction provided by the Ruc2 flow can also be considered as an intermediate position between the reverse to normal polarity change. These observations leads to consider that at least the Mont des Ruches section recorded all the reversals. Some doubts exist for the Mont de la Rabouillère and Mont de la Tourmente sections. Their intermediate VGP directions could signal unrecorded reversals, and thus the eruption rates calculated on these sections may be exaggerated. Finally, the estimated palaeosecular variation substantiates an axially symmetric palaeomagnetic field with equatorial symmetry since they are coherent with the Model G of palaeosecular variation proposed for the Oligocene period (McFadden et al., 1991).

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